

Lecture 2.3 – Reading and Writing Files

Specific Learning Objectives:

3.1 – Load, clean, and organize data in R.

Using Spreadsheets Safely

- I probably can't convince you to not use spreadsheets, so you may as well use them SAFELY (so you can't upload the data to R later).
- Align your data in the upper left, leaving no extra columns on either side.

DON'T 🙅

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6			Blue Crab Data						
7									
8			Re	Rep	Orient	current	Model	Speed	
9			1.59	1	B	0	C	0.43839515	
10			1.59	2	B	0	C	0.51984394	
11			1.59	3	B	0	C	0.4109238	
12			1.83	1	B	4	C	0.5698963	
13			1.83	2	B	4	C	0.72409106	
14			1.83	3	B	4	C	0.57350835	
15			2.82	1	B	15	C	1.10059978	
16			2.82	2	B	15	C	1.53748625	
17			2.82	3	B	15	C	1.27945825	
18			0.57	1	B	0	C	0.08905971	
19			0.57	2	B	0	C	0.12984493	
20			0.57	3	B	0	C	0.07397596	
21			0.23	1	B	4	C	0.04501641	
22			0.23	2	B	4	C	0.03796072	
23			0.23	3	B	4	C	0.04804528	
24			0.94	1	B	15	C	0.16302115	

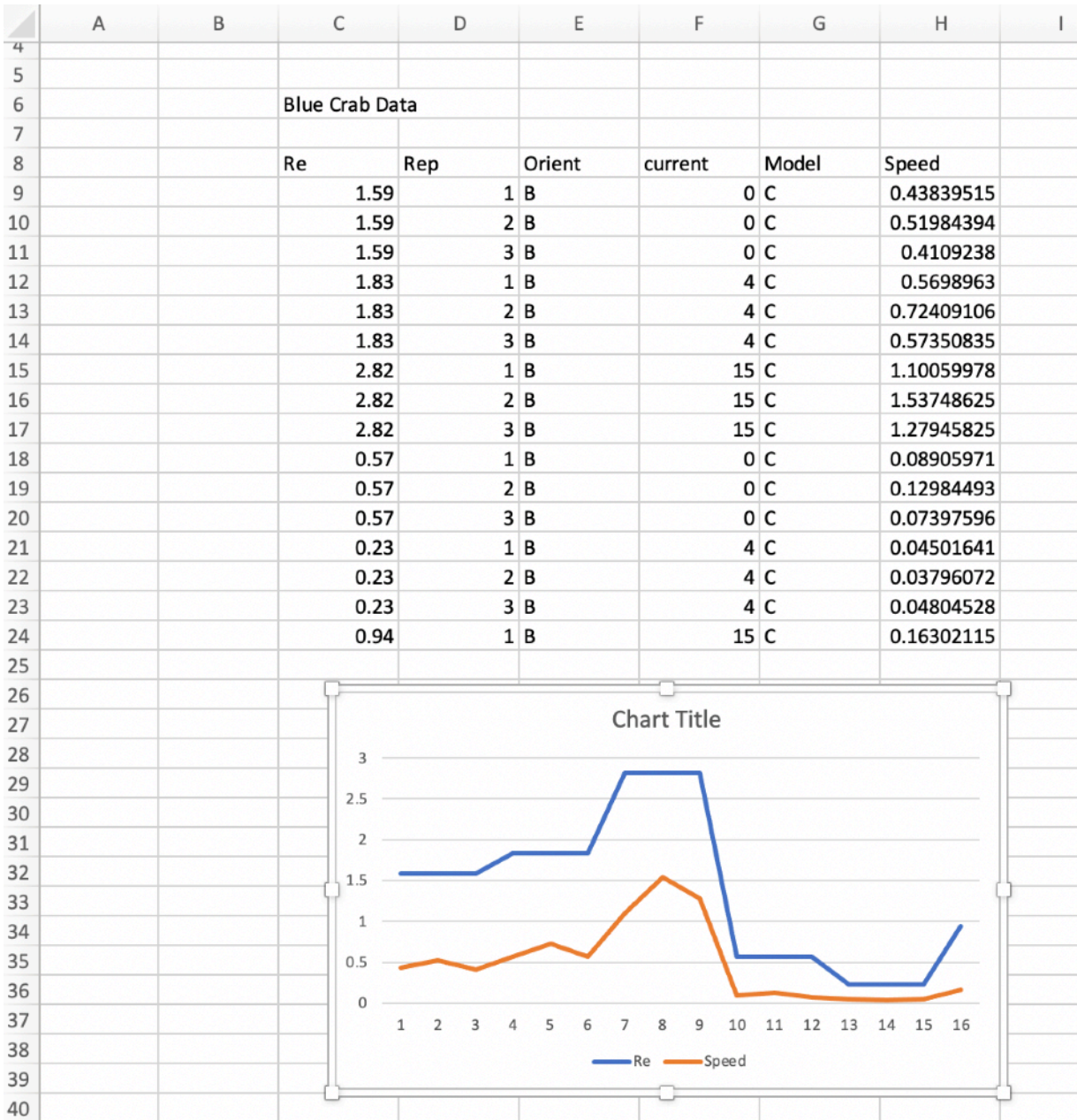
DO 👍

	A	B	C	D	E	F	G
1	Re	Rep	Orient	current	Model	Speed	Refresh
2	1.59	1	B	0	C	0.43839515	
3	1.59	2	B	0	C	0.51984394	
4	1.59	3	B	0	C	0.4109238	
5	1.83	1	B	4	C	0.5698963	
6	1.83	2	B	4	C	0.72409106	
7	1.83	3	B	4	C	0.57350835	
8	2.82	1	B	15	C	1.10059978	
9	2.82	2	B	15	C	1.53748625	
10	2.82	3	B	15	C	1.27945825	
11	0.57	1	B	0	C	0.08905971	
12	0.57	2	B	0	C	0.12984493	
13	0.57	3	B	0	C	0.07397596	
14	0.23	1	B	4	C	0.04501641	
15	0.23	2	B	4	C	0.03796072	
16	0.23	3	B	4	C	0.04804528	
17	0.94	1	B	15	C	0.16302115	
18	0.94	2	B	15	C	0.20110097	
19	0.94	3	B	15	C	0.15687138	

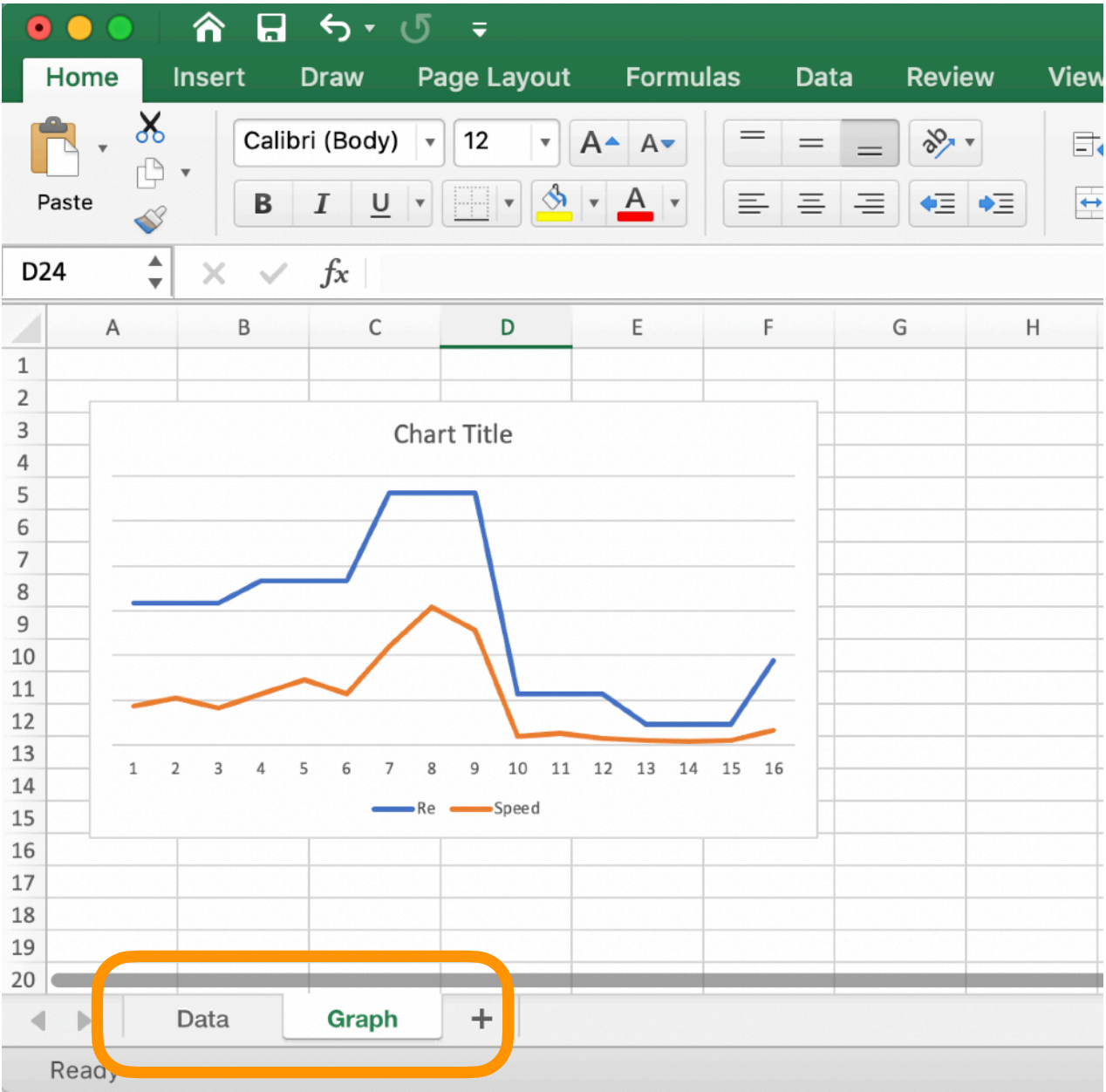
Using Spreadsheets Safely

- Put objects like charts and graphs on a separate sheet from the data.

DON'T 🙅



DO 👍



Using Spreadsheets Safely

- Only put one data set per sheet.

DON'T 🙅

J6 x ✓ fx Baby crab data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
2																	
3																	
4																	
5																	
6			Blue Crab Data							Baby crab data							
7																	
8			Re	Rep	Orient	current	Model	Speed		number	size (mm)	log size	sex	Hair diameter	Hair length	Cuticle thickness	
9			1.59		1 B		0 C	0.43839515		1	27	3.29583687	m	8.10E-06	648	7.68E-07	
10			1.59		2 B		0 C	0.51984394		2	8.5	2.14006616	f	6.73E-06	465	7.19E-07	
11			1.59		3 B		0 C	0.4109238		3	11	2.39789527	f				
12			1.83		1 B		4 C	0.5698963		4	5	1.60943791	f	5.68E-06	374		
13			1.83		2 B		4 C	0.72409106		5	11.5	2.44234704	m	7.44E-06	484	8.74E-07	
14			1.83		3 B		4 C	0.57350835		6	28	3.33220451	m	8.49E-06	660	1.01E-06	
15			2.82		1 B		15 C	1.10059978		7	25	3.21887582	f	8.01E-06	543	1.01E-06	
16			2.82		2 B		15 C	1.53748625		8	15	2.7080502	m	7.00E-06	505	7.04E-07	
17			2.82		3 B		15 C	1.27945825		9	22	3.09104245	f	8.81E-06	552		
18			0.57		1 B		0 C	0.08905971		10	4.9	1.58923521	m	5.69E-06	347	6.39E-07	
19			0.57		2 B		0 C	0.12984493		11	5.85	1.76644166	m	5.16E-06	356		
20			0.57		3 B		0 C	0.07397596		12	7.12	1.96290773	m	5.53E-06	400	6.31E-07	
21			0.23		1 B		4 C	0.04501641		13	16	2.77258872	m	8.38E-06	435	7.91E-07	
22			0.23		2 B		4 C	0.03796072		14	13	2.56494936	f	7.75E-06	468	5.93E-07	
23			0.23		3 B		4 C	0.04804528									
24			0.94		1 B		15 C	0.16302115									
25																	
26																	
27																	

DO 👍

D19 x ✓ fx

	A	B	C	D	E	F	G	H	I
1	number	size (mm)	log size	sex	Hair diameter	Hair length	Cuticle thickness		
2	1	27	3.29583687	m	8.10E-06	648	7.68E-07		
3	2	8.5	2.14006616	f	6.73E-06	465	7.19E-07		
4	3	11	2.39789527	f					
5	4	5	1.60943791	f	5.68E-06	374			
6	5	11.5	2.44234704	m	7.44E-06	484	8.74E-07		
7	6	28	3.33220451	m	8.49E-06	660	1.01E-06		
8	7	25	3.21887582	f	8.01E-06	543	1.01E-06		
9	8	15	2.7080502	m	7.00E-06	505	7.04E-07		
10	9	22	3.09104245	f	8.81E-06	552			
11	10	4.9	1.58923521	m	5.69E-06	347	6.39E-07		
12	11	5.85	1.76644166	m	5.16E-06	356			
13	12	7.12	1.96290773	m	5.53E-06	400	6.31E-07		
14	13	16	2.77258872	m	8.38E-06	435	7.91E-07		
15	14	13	2.56494936	f	7.75E-06	468	5.93E-07		
16									
17									
18									

BlueCrabData BabyCrabData +

Using Spreadsheets Safely

- Use plain text and avoid special characters and inserting things like pictures, clipart, and video.



- If you need to write notes, make a “comments” column and restrict text to that column. Don’t write comments in random cells!
- If a column should be numeric, make sure ALL data is numeric in that column.
- Any calculations should (ideally) go in another sheet.

Check Your Understanding

Import the data set in `tomatohaul2021.csv` as `tomato.data`. The first column should be row names, the dates should be characters, and there is a header.

Check Your Understanding

Use the pipe operator to create `babycrab.clean` object that takes `babycrab.data` and cleans it with each of the cleaning steps at once but in this order:

1. cleans column names
2. removes empty rows and columns
3. removes all rows with NA values

Check Your Understanding

Pick a data frame from `data()` and save it as a csv file in the `data/` folder.

In-class Exercises

1. **Work on Assignment 2.2 using R Markdown.**

Action Items

- 1. Complete Assignment 2.2 using R Markdown.**
- 2. Read Davies Chapter 8 and Chang Chapters 1-2 for next time.**